

Mme Yue CAO

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EDUCATION

09/2025 – present **MASTER** Natural Language Processing (NLP) – Second year

Institut National des Langues et Civilisations Orientales (INALCO) – PluriTAL

Main courses:

- **NLP engineering:** including Machine Learning, Neural Networks; Web technologies, algorithmic programming; Object-oriented programming
- **Linguistics:** Linguistics for multilingual NLP; Language Modeling

09/2024 – 06/2025 **MASTER** Natural Language Processing (NLP) – First year

Université Paris Nanterre – PluriTAL

Main courses:

- **NLP and Linguistic Engineering:** including Programming; Corpus processing tools; Textual statistics; Mathematics; Parallel and comparable corpora
- **Linguistics:** Treebanks and parsing; Introduction to text mining; NLP English, etc.
- **Digital document:** Structured document ; Databases – SQL

09/2020 - 06/2024 **BACHELOR** Letters, French Language and Literature

Heilongjiang International University, Heilongjiang, China

Dissertation :

Analyse des méthodes de traduction des métaphores en langue familière :
l'exemple du feuilleton *Dix pour Cent*

PRACTICAL PROJECTS

• Preprocessing and statistical analysis of real-time writing data (annotated corpus)

Study of the relationship between real-time writing dynamics (pauses, deletions, modifications), with chunk segmentation via [SEM](#) and it extracts syntactic information, revealing an interaction between linguistic organization and writing dynamics. [\[Gitlab\]](#)

• Pipeline RAG

Developed a complete RAG pipeline. Built a mini RAG system using ChromaDB and Gemini 2.5 for question answering about the characters base a raw text novel *My Brilliant Friend*.

• Replication-based extension of DEFT'09 Task 3

Modified the original Task 3 setup by adding a de-duplicated test evaluation and creating an alternative training corpus via rule-based subjective-sentence extraction, then training tuned sklearn classifiers on TF-IDF word n-gram features.

• Training Transformer models for genre classification and classical Chinese generation

Training from scratch of a genre classifier and generative models with two Transformer architecture (BERT/GPT-type) to identify the genre of a text and generate classical Chinese, experimentation on historical prose and Tang poetry.

- **CIFAR-10 Image Classification with a CNN**

Train a CNN on CIFAR-10 with PyTorch and evaluate it. Then improve performance by tuning the architecture and training (data augmentation, mini-batches+shuffle and ReLU) setup to achieve the best possible results on CIFAR-10.

- **Fine-tuning an ASR Model for Chinese Dialect Recognition and Transcription**

A baseline is generated using funasr/paraformer-zh without any preprocessing. Based on the baseline results, corresponding methods are applied for fine-tuning, ultimately achieving a satisfactory result.

Other projects are in progress but not added yet.

TECHNICAL SKILLS IN NLP

Python : pandas, NumPy, scikit-learn, NLTK, spaCy, Stanza, THULAC, Word2Vec; Deep Learning : Keras, PyTorch; Corpus : HTML, XML, CSV, JSON; SQL and Bash/Shell

Deep Learning: CNN, RNN (LSTM, GRU), Seq2Seq+Attention, Transformer

Software and tools : VS Code, Jupyter Notebook, Oxygen, XML, MySQL, Unitex, Praat, marimo, Overleaf, Tesseract5

Tasks already done : classification de sentiments, génération, extraction, synthèse vocale, ASR, TTS, OCR, RAG

LANGUAGES

- Chinese (native language)
- French (B2/C1)
- English (B2/C1)

QUALITIES

- Good teamwork skills
- Curiosity and initiative
- Patient, stable and calm